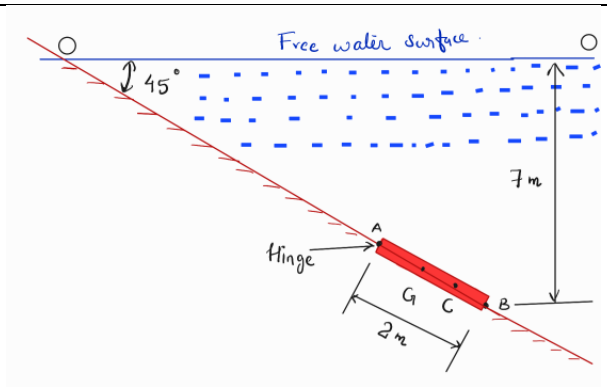
MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF AERONAUTICAL & AUTOMOBILE ENGINEERING			
END SEMESTER EXAMINATION		**NOV 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Aeronautical Engineering**	**Course Type**	**C**
Course Name	**Fluid Dynamics**	**Course Code**	**AAE 2124**
Semester	**III**	**Max Marks**	**50**
Date	**25/11/2024**	**Duration**	**3 hours**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET		BL
			CO	PO	CLO	AHEP4 LO	
1A	A solid cube with sides 0.5 m each is made of a material of specific gravity 0.5. The cube floats in a liquid of specific gravity 0.95 with two of its faces horizontal. Examine its stability.	**3**	CO3	PO1, PO2, PO3	CLO3	C1, C2, C3	5
1B	Briefly explain the following terms: i. Hydrostatic law ii. Piezometer iii. Absolute pressure iv. Dynamic pressure	**4**	CO1	PO1	CLO1	C1	2
1C	Derive an expression for the capillary rise at a liquid having surface tension σ and contact angle θ between two vertical parallel plates at a distance W apart. If the plates are of glass, what will be the capillary rise of water having σ = 0.073 N/m, θ = 0°? Take W = 1 mm.	**3**	CO1	PO1	CLO1	C1	6
2A	An inclined rectangular sluice gate AB 2 m x 6.5 m in size as shown in the figure is installed to control the discharge of water. The end A is hinged. Determine the normal force to the gate applied at B to open it.	**5**	CO3	PO1, PO2, PO3	CLO3	C1, C2, C3	5

							
2B	For a flow field of fluid is given by $V = xy\,i + 2yz\,j - (yz + z^2)k$ i. Show that it represents a possible case of flow ii. Check if the flow is rotational or not. iii. If rotational, find the angular velocity, and vorticity at the point (2,4,6)	**3**	CO2	PO1, PO2	CLO2	C1, C2, C3	3
2C	The velocity components in a three-dimensional flow are $u = x^2 + y^2z^3, v = -(xy + yz + zx)$ Determine the missing component of velocity such that the continuity equation is satisfied.	**2**	CO2	PO1, PO2	CLO2	C1, C2, C3	3
3A	With a schematic, explain the working of a Pitot static probe.	**4**	CO2	PO1, PO2	CLO2	C1, C2, C3	2
3B	A horizontal Venturimeter with an inlet diameter of 200 mm and a throat diameter of 100 mm is used to measure the flow of water. The reading of the differential manometer connected between the inlet and the throat is 180 mm of mercury. If the coefficient of discharge is 0.98, determine the flow rate of the flow.	**3**	CO2	PO1, PO2	CLO2	C1, C2, C3	3
3C	Derive the Euler's equation of motion for a fluid flow.	**3**	CO2	PO1, PO2	CLO2	C1, C2, C3	6
4A	The discharge Q of a centrifugal pump depends on the mass density of the fluid (ρ), the rotational speed of the pump (N), the diameter of the impeller (D), the manometric head (H), and the viscosity of the fluid (μ). Show that $Q = ND^3\,\varphi\left(\frac{gH}{N^2D^2}, \frac{\mu}{\rho ND^2}\right)$	**5**	CO4	PO1, PO2, PO3	CLO4	C1, C2, C3	6
4B	Analyze the differences between laminar and turbulent flow in a pipe regarding their velocity profiles, energy losses, and flow behavior.	**3**	CO4	PO1, PO2, PO3	CLO4	C1, C2, C3	4
4C	Describe the different types of losses that occur in pipe flow.	**2**	CO4	PO1, PO2, PO3	CLO4	C1, C2, C3	2

5A	A lubricating oil of viscosity 1 poise and specific gravity 0.85 is pumped through a 30 m diameter pipe. If the pressure drop per unit length of the pipe is 25 kN/m^2, evaluate: i. Mass flow rate in kg/s ii. Centre-line velocity iii. Shear stress at the pipe wall iv. Frictional force on the pipe wall for a length of 50 m. v. Power required to maintain the flow in the 50 m length of the pipe. vi. Velocity and shear stress at 10 mm from the wall of the pipe.	**5**	CO4	PO1, PO2, PO3	CLO4	C1, C2, C3	5
5B	Briefly describe the development of a boundary layer over a flat plate, with various flow regions using a schematic.	**3**	CO5	PO1, PO2, PO3	CLO5	C1, C2, C3	2
5C	Explain the three ways of expressing the boundary layer thickness over a flat plate.	**2**	CO5	PO1, PO2, PO3	CLO5	C1, C2, C3	2

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***7***
2	***16***
3	***7***
4	***15***
5	***5***

BL	Total Marks
2	**15**
3	**09**
4	**03**
5	**14**
6	**09**